Updated: Gomes June 2025

% C on Elemental Analyzer Protocol

<u>Goal: To Measure the % carbon in the soil samples with depth and across the field (rep and treatment) at the beginning of each year of the field trial</u>

<u>-> to determine % organic vs. inorganic C</u>

1. Measuring Sept soils from both years
    a. Each treatment, depth, and replicate (48 bags * 2 years) =96 samples)
2. Air dry soil samples (lay out on clean paper on a shelf in the lab) (paper towels can have carbon)
3. Grind soil sample with ball mill in Mitchell (avoid any cross contamination)
4. Order silver capsules for the soil tins (5x9s)
5. Weigh out the first set of soil samples into capsules: record 3 decimal points; this is for **total C** (tin capsules), close into balls, (20-35mg) (if mass is too large can get sample stacking inside column and get further from the hot spot in the column and introduce analytical drift)

For the acidified samples in Dave's Lab: to remove the inorganic C from the soil

6. To start, Weigh out a **subset** of Yr1 and Yr2 samples (8 from each, mainly 60cm and 90cm depth) for organic C (this will be the soil you acidify to remove the inorganic C) (silver capsules 5x9) ~15mg; keep capsules open so can add acid
7. After weighing out (record mass on Dave's microbalance) soil into silver capsules..
8. Add 40? ul of the 1M HCl and see if there is any effervescence/bubbling in the microscope (don't overfill the capsule!! Determine after first set of samples)
9. Then let the samples sit on a hot plate (at 50C) for a 1 hour so the samples react and the acid evaporates so that the capsules don't overflow. Check again for bubbles in the microscope! Keep the sample wet with HCl .
10. <u>If there is no bubbling</u>, then you do one more addition of acid **(+1)** and check for bubbling, then they dry on the hot plate for 2? hours, and then they dry for 24 hours in a drying oven (35-40C).
11. I<u>f there is bubbling</u> then you would repeat the hot plate drying and then do another acid addition and check for bubbling again.
12. Record how many additions for each sample on the printed sheet!
13. Because the method is based on if the samples effervesce rather than adding a particular amount of acid, it's good to have internal control samples throughout the run to ensure that your acidification is consistent/effective.

14. Run both sets of foil balls (total C and organic C) on the EA: silver and tin capsules separately

15. Calculate % organic and % organic C for each date/depth/treatment/replicate combination (then calculate the %inorganic C)

Total %C= %organic C + %inorganic C

<u>Starting the Elemental Analyzer</u>

Start by turning on the desktop computer, the password is located behind the computer.

To start a sample run, check the diagnostics, the temperature should reach around 950 and 850 degrees in the furnaces

Make sure that the method on the home screen is the correct one, so the data saves to the correct file. It should be the method in the Anna_Gomes folder, double check and adjust if necessary.

After checking the diagnostics, begin setting up the run tables, press the blue icon to the right of the green start button that reads diagnostic to have the machine begin warming up

To start the run table, delete all information from the previous run, and place a bypass sample as the first sample, this is filled with an unweighed amount of aspartic acid, the next sample should be a blank tin (nothing in it)

Run three blanks each with the acid located to the right of the tin weighing station, it is wise to weigh samples while the machine is warming up, that process takes roughly 30 minutes

Label the standards std1, std2, std3 with approximately 5mg, 10mg, 15mg of standard respectively

In the sample table, insert one completely empty line- the machine will pause here, check the standard curve, then delete the line and restart the run

When filling the tins, 16-18 mg is usually a safe bet, for agricultural soils 10-12 mg should suffice, and for plants 2-4 should be good (compost ~ 10mg). If using the new equipment, weigh and tare the tin, add your sample then mark the weight. Then, place the tin in the divet located on the silver cylinder. Fold the sides in, to flatten, use the silver stick, and set in a sample tray. Weigh to three decimals.

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When labeling the samples, it is best to include the slot number in the name, to make sure the samples and data match; you can not directly type into the sheet, so the sample names will auto-populate depending on the number of samples you enter into the beginning of the sheet

To place the samples inside the slots, carefully lift the lid off of the turning table, and place it upside down on a paper towel on top of the machine. Then, slowly load samples into their corresponding slot numbers. Begin with one, the machine only rotates in one direction, clockwise. Slowly turn the slots, aligning the silver square to the orange drop-off first, and then turning to adjust the numbers so that 1 (one) is the first slot. This is an important step, data will not process correctly if it is misaligned

Once the sample run list has been completed and all samples have been loaded, press the green start run button, if running early in the day (as is preferred) do not check any of the boxes, simply click the start now button/.

Listen and look to make sure that your first sample has dropped. The first sample was a bypass, so there should be no live data analysis. To watch the live data analysis, click the view button on the home screen. The first sample being analyzed should be sample two, and there should be two peaks that begin to form, carbon then nitrogen. Each sample takes approximately seven minutes, after the blanks come back, check to see if they are good. Once those have begun, the EA will pause and then you can start again to run your samples.

Post-Sample Run/Data Processing:

After your samples have finished running, click on the summary table. Check that the calibration curves look correct and are acceptable. Then, click on File and click Export Data. Export the data onto an Excel spreadsheet, be sure to save it to the correct folder. Then, transfer that Excel spreadsheet onto a flash drive.

If you are finished running the samples, click on the same diagnostic logo you did earlier. There, click send machine to standby. Once the machine colors standby, turn off the computer, and do not save the method.

There is no need for any sample clean-up, once in a while the ashtray will fill up, however, that is a job for Taylor.

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8/12/25 Foil Balling Demo with Taylor
EA: 32 samples at a time

Overview:
Get 4 runs worth of samples before starting EA
20 runs on EA total (including the organic ones)
2 runs in a day, 4 hours per run

4 standards, 1 blank, 1 bypass
26 actual samples, 6 standards
2 runs in a day, can just do one set of standards

Taylor has sample template tables
Can make a standard tray
Good bookkeeping is very important!!!!
Trays are in EM1
- Lids just lock in place, need to tape shut
- 25 mg of Sample (around 1mg C) is our target
- Better to go under rather than over
- For organic, 10-15mg
- Sample 25 mg, + or - 10, record exact mass on sheet (at least 2 decimal places)
- Can weigh out 10 sets of standards to have as reserve
- Use bigger tins
- Need to get silver tins to replace Taylors, he has a few 100 !

MAKING FOIL BALLS:
Getting Started:
- Wash all tools with isopropanol at the beginning of each day (all weighing out tools, etc)
- Tin blank: crush up tin into a ball
- Standard 1: tare empty tin, add ~1mg (or whatever standard you are aiming for), weigh and record
- (if you overshoot, use it for one of the higher standards, easier to add sample than to take away)

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- Record tray position
- Pinch shut when taking out of the tray, fold over top, fold the sides in until it's a bit smaller, squish with paddle shaped one (careful not to tear the capsule!)
- If you see powder, throw away and start over (clean off surface)

Running EA: DEMO WITH TAYLOR 10/22 3pm

Can run 63 samples at a time

1. Log on, NC
2. Takes 15-20 mins to get out of stand-by and heat up
3. Edit, edit EA parameters, uncheck and send
4. View, view EA status
5. Load temp method file to prevent saving over someone else's file
6. Load new method
7. flash data/one temp/one temp method
8. Create folder and file specific to today's run
9. Method file tells where files will be saved
10. Create data sheet
11. Enter blanks and standards on data sheet manually
12. Skip bypass on second set
13. 52 actual samples per run
14. There is no save, be careful! Ok saves it but then have to go back
15. Plastic piece on top, MAKE SURE ITS LINED UP
16. Run bypass sample
17. Force to standby